



When Nmap Says Nothing: A Small Change That Made Internal Service Discovery Possible During an...

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During internal penetration tests, it is common to focus on exploiting vulnerabilities, identifying misconfigurations, and assessing privilege escalation paths. However, some of the most valuable lessons come from the reconnaissance phase, particularly when tools do not behave as expected.

During a recent authorized Active Directory security assessment, I encountered a situation where a standard Nmap scan was not producing the results I expected. The target system was known to be online, but the scan was not identifying services correctly. After investigating the issue and adjusting the scan configuration, I was able to successfully enumerate the target and continue the assessment.

This article explains what happened, why it happened, and what security professionals should understand about Nmap's behavior on internal networks.

The Initial Scan

The first command I used was

```
nmap -sV -Pn <ip-address>
```

This is a fairly common command during internal assessments.

The options perform the following functions:

- `-sV` enables service version detection.
- `-Pn` tells Nmap to skip host discovery and assume the target is online.

Since the target was already known to be reachable, using `-Pn` appeared to be the correct choice. However, the scan did not provide the information I expected.

At first glance, this can be confusing. Many penetration testers assume that `-Pn` completely disables host discovery mechanisms and forces Nmap to begin scanning immediately. While that is partially true, there is an important detail that often gets overlooked.

Understanding ARP on Local Networks

When Nmap scans a host located on the same local subnet, it behaves differently than when scanning a remote system across routers.

Before Nmap can send TCP packets to a local host, it must determine the target's MAC address. To do that, it uses the Address Resolution Protocol (ARP).

A simplified example looks like this:

```
Who has 192.168.1.100?  
Tell 192.168.1.50
```

The target responds with its MAC address, allowing communication to continue.

This process is normally fast and transparent. Most of the time, users never notice it.

The important point is that even when `-Pn` is specified, Nmap still prefers ARP-based discovery on local Ethernet networks.

Many security professionals assume that `-Pn` completely eliminates all discovery mechanisms. In reality, ARP is still used under certain circumstances because Nmap considers it the most reliable method for determining whether a local host exists.

Why This Can Become a Problem

In corporate environments, especially those containing Active Directory infrastructure, security controls are often more sophisticated than many people realize.

Depending on the environment, ARP traffic may be:

- Monitored by network security tools
- Restricted by segmentation controls
- Filtered by endpoint security products
- Interfered with by virtualization or network configurations
- Logged by security monitoring systems

When this occurs, ARP-based discovery may not behave as expected.

As a result, Nmap may struggle to properly identify hosts before beginning the actual service scan.

This can create the impression that the target is unavailable even when it is online and responding to traffic.

The Command That Worked

After troubleshooting the issue, I modified the scan and used the following command:

```
nmap -sV -Pn --disable-arp-ping --packet-trace -D RND:5 <ip-address>
```

This immediately produced the results I was looking for and allowed service enumeration to continue successfully.

Let's examine the additional parameters and understand why they mattered.

The Most Important Switch: — disable-arp-ping

The key difference was the addition of:

```
--disable-arp-ping
```

This option instructs Nmap not to perform ARP-based host discovery.

Instead of relying on ARP requests to determine whether the target exists, Nmap proceeds directly with the scan packets.

In my case, this appears to have been the deciding factor.

Once ARP discovery was disabled, Nmap stopped depending on local ARP responses and began sending probes directly to the target. Service detection then worked as expected.

For penetration testers operating in monitored enterprise environments, understanding this behavior can save a significant amount of troubleshooting time.

Using — packet-trace for Visibility

I also enabled:

```
--packet-trace
```

This option prints every packet sent and received during the scan.

For example, it allows you to see:

- TCP SYN packets leaving your machine
- Responses from the target
- Retransmissions
- Connection attempts
- Firewall behavior

During troubleshooting, this visibility can be extremely valuable.

Instead of guessing whether packets are reaching the target, you can observe the communication process directly.

When assessing internal networks, packet tracing often helps identify whether the problem is related to routing, filtering, firewalls, endpoint controls, or host responsiveness.

What About the Decoys?

The final addition was:

```
-D RND:5
```

This option generates five random decoy addresses.

The idea is to make the scan appear as though it originates from multiple systems rather than a single source.

Historically, decoy scanning was used to complicate attribution by mixing the real scanning host among several fake addresses.

While the technique still functions, modern enterprise environments typically maintain extensive logging capabilities.

Security teams often have access to:

- Firewall logs
- Switch logs
- Endpoint detection systems
- Network flow records
- Authentication events

Because of this, decoys should not be viewed as a method for bypassing enterprise monitoring. Their practical value today is generally lower than it was years ago.

In this assessment, the decoy option was not responsible for the successful service enumeration. The real benefit came from disabling ARP-based discovery.

Key Takeaways for Penetration Testers

This experience highlighted several lessons that are worth remembering during internal security assessments.

First, `-Pn` does not completely eliminate all host discovery mechanisms. On local networks, ARP can still play an important role.

Second, when a target is known to be online but Nmap is not behaving as expected, it is worth considering how ARP discovery may be affecting the scan.

Third, troubleshooting options such as `--packet-trace` can provide valuable visibility into what is actually happening on the network rather than relying on assumptions.

Finally, understanding how tools operate under the hood is just as important as knowing how to run them. Penetration testing is not simply about executing commands. It is about understanding the protocols, network behavior, and security controls that influence the results.

Final Thoughts

Many penetration testers use Nmap every day, yet relatively few spend time examining the underlying mechanisms that make the tool work.

This assessment served as a reminder that small configuration changes can have a significant impact on results. What initially appeared to be a scanning issue was ultimately related to how Nmap handled local network discovery.

The next time a known host refuses to cooperate during an internal assessment, it may be worth taking a closer look at ARP before assuming the problem lies elsewhere.

Sometimes the difference between seeing nothing and seeing everything is a single switch.

Let's Connect

If you found this useful or want to discuss cybersecurity, penetration testing, or security research, feel free to connect with me.

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